



Powering Efficiency in Farming & Oilfield Operations

Unique technology for carbon, cost and energy reduction with guaranteed energy savings



INNOVATIVE ENERGY AND
CARBON REDUCTION
SOLUTIONS



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ABOUT US

Engineering energy efficiency that pays off.

Emissis North America, LLC, based in Las Vegas, Nevada, is the U.S. subsidiary of Emissis (Electrical Mechanical & Cooling Ltd). Emissis is a specialist engineering firm with over 20 years of experience in delivering high-performance mechanical and electrical solutions. We provide independent, critical power and cooling services tailored to industrial and commercial environments across the US.

Being independent means we're not tied to specific products - we choose the right technologies to meet your site's exact needs. Backed by a skilled in-house engineering team, we're known for our technical rigor, compliance expertise, and project efficiency.

Our Mission

To help clients reduce energy costs, extend asset lifespan, and accelerate carbon reduction - without disrupting operations.

Our Core Technology: enPact

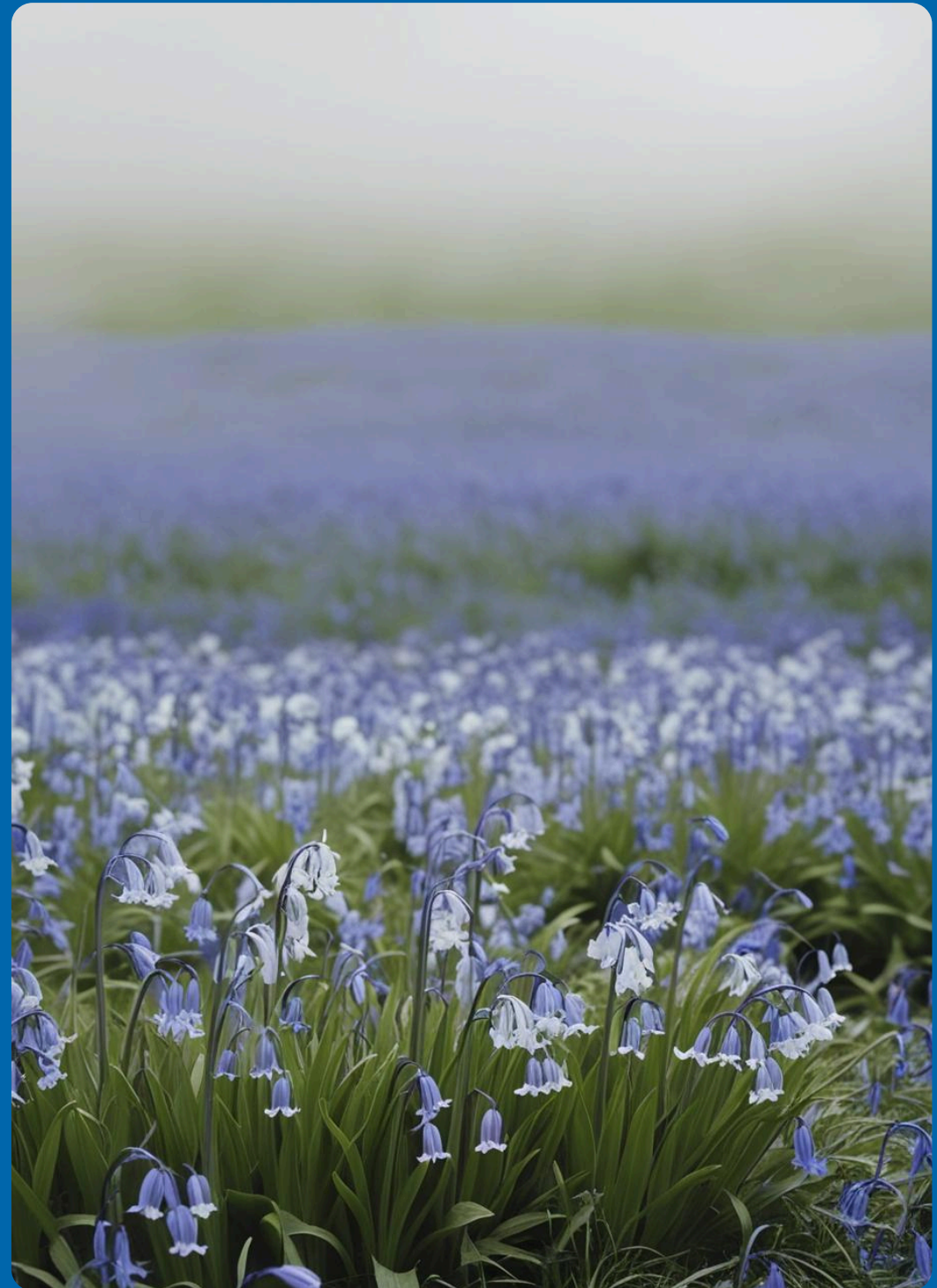
We deliver a suite of globally patented energy-saving solutions under the enPact brand, designed to optimize energy use across key systems:

- enPact Refrigeration & Cooling
- enPact Chillers
- enPact Electrical Optimization
- enPact Energy & Carbon Management
- enPact Engineered Solutions

We serve a wide range of sectors including:

- Kitchens, Bars & Restaurants
- Food Retail & Production
- Heavy Industry & Manufacturing
- Hospitality & Built Environment
- Farming & Oilfield Operations

From initial survey through to installation, monitoring, and payback verification - we deliver full-service energy optimization that reduces cost and carbon for some of North America's most recognized businesses.



THE CHALLENGE

Farming - Irrigation & Pivot Units

- Pivot irrigation systems are powered by large inductive motors.
- A single system can consume electricity costs of up to US\$15,000/month.
- Inefficiencies in power quality, harmonics, and voltage imbalance drive unnecessary energy losses.
- Wasted energy = higher operating costs, reduced equipment lifespan, and unnecessary carbon emissions.



Oilfield - Pumping Units

- Pumpjacks and electric pumping rigs (particularly within the oilfield regions of North America) often consume up to US\$50,000/month per unit.
- Similar inductive loads create inefficiencies that are not visible on standard electricity bills.
- Even a small percentage of savings translates into tens of thousands of dollars per year per pump.
- Operators face growing pressure to improve cost efficiency and reduce environmental impact without disrupting operations.



HOW IT WORKS:

- FilterPro produces savings using Spintronics technology.
- Spintronics uses a Ferromagnetic layer to align spin orientation and hence reducing I²R losses in a circuit.
- The unit is a fast fit device connected in parallel to the target circuit.
- Higher savings achieved on Inductive circuits (motors, compressors, pumps, welders etc).
- Parallelable so can cover large incoming supplies.
- Savings are visible within 2 to 5 days from installation. The initial time is required to align spin orientation.
- Supplied with a 5 year warranty.



enPact[®] FilterPro

Save 6-15% on High Induction Load Electricity Costs

enpact FilterPro can be connected on the incoming supply covering the whole installation or targeted to individual large inductive loads.

Has no impact to Voltage or Torque aspects. Eliminates any concerns to operation aspects.

Peak inrush currents are greatly reduced hence reducing Kw consumption which in turn reduces carbon footprint. FilterPro with its proven technology delivers savings of 6 to 18%

Proven Results

Across industrial and commercial sites worldwide, FilterPro has consistently delivered:

- ROI typically within 9-12 months.
- Significant carbon reduction aligned to ESG and regulatory requirements.
- Extended asset life due to reduced electrical and thermal stress.





enPact FilterPro[®]

CASE FOR ACTION

Why Now?

- Energy prices remain volatile, and operators in both farming and oil are highly exposed.
- Power losses are invisible costs that compound every month.
- FilterPro offers a proven, low-risk solution with measurable results.

How Savings Are Measured:

- Initial site survey and load analysis.
- Installation of FilterPro units.
- On-site metering installed to baseline performance and verify savings.
- Transparent reporting provided to client, proving ROI.



By installing enPact FilterPro, operators in farming and oil can achieve immediate, verifiable reductions in energy costs, improve operational reliability, and contribute to sustainability goals.



Our EnOS platform integrates on-site operational technology with cloud-based intelligence to provide real-time energy data, along with data-driven carbon monitoring, reporting, and abatement solutions.

Energy Monitoring

- Monitor your energy consumption in real-time with our advanced IoT technology.
- Identify energy conservation measures utilizing the capabilities of artificial intelligence.
- Receive alerts in the event of overconsumption in any of your showrooms.

Carbon Reporting

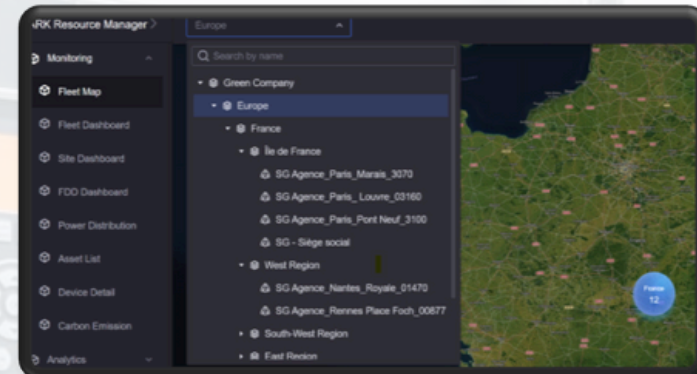
- Utilize real-time data to assess, benchmark, and report CO2 emissions both at individual sites and across the entire portfolio.
- Strategically plan and prioritize abatement initiatives based on these assessments.

Savings Identification

- Analyze actual consumption data in comparison to your established energy baseline to identify discrepancies.
- Integrate a real-time AI module to monitor and report on energy savings.
- Analyses site-level performance to detect inefficiencies and deliver data-driven optimization strategies to reduce energy - generating on average an additional potential saving of 25%.

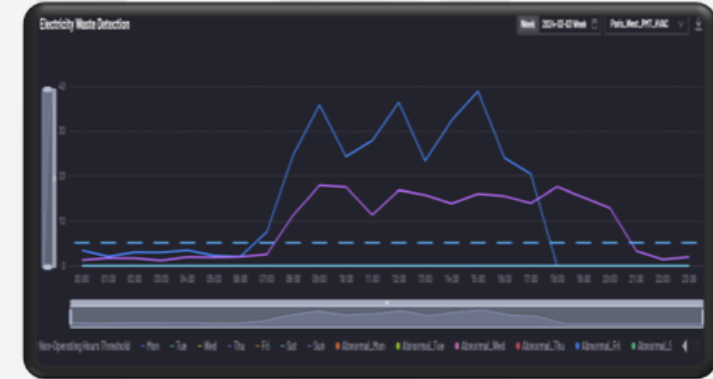
Track Performance Against Targets

- Establish energy and carbon targets in alignment with the Science Based Targets initiative (SBTi) and monitor the performance of each facility relative to these objectives.



**Site by Site View
Portfolio View
Asset View**

**Customizable site dashboard:
aggregated energy consumption data**



Set up energy performance KPI's based on the site's specifics



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